BT 101
Mid term question
Dec2017

1: Write short note on population dispersion ?5

The ways in which populations are spread across Earth has an effect on the environment. Developing countries tend to have higher birth rates due to poverty and lower access to family planning and education, while developed countries have lower birth rates. In 2015, 80 per cent of the world's populations live in less-developed nations. These faster-growing populations can add pressure to local environments

The Three Main Types of Population Dispersion

Population dispersion is the observation of where individuals are found in a habitat. How individuals "disperse" themselves. There are three main types of dispersion: **clumped**, **uniform** and **random**.

Clumped Dispersion

Is the tendency for populations to be found in tight clusters, dispersed across a large landscape. In between these population hubs, very few to no individuals are usually found. This sort of a dispersion can be caused by a number of factors. Some species cluster together for protection, while others group around natural resources necessary to their survival.

Uniform Dispersion

Is the tendency for populations to be found evenly distributed about their habitat. This is generally caused by a species ability to survive anywhere in their habitat - they use the resources found immediately around them, and spread out as to use all of the available resources.

Random Dispersion

Is the tendency for populations to be found randomly about their habitat. In immobile species, this is usually caused by their ability to live anywhere in a given habitat, except, they are limited to growing wherever they are first set root (which is usually caused randomly, from spores drifting in the wind to seeds falling and tumbling on the ground). In motile populations, individuals are able to move about their habitat, so that at any given instance, they can be found anywhere about their environment.

2: Write short note on Soil horizone?5

>A vertical section of soil that tells about various layers of the soils as zones. These zones are named as soil horizons.

In a typical soil profile, four zones are exposed,

1. **>O-horizon**
2. **A-horizon**
3. **B-horizon**
4. **C-horizon**

>O-horizon

Uppermost zone of soil surface where fallen leaves and other plant matter is deposited.

A-horizon

Zone of leaching because percolating rain picks up the soluble materials and transport it downward to B-horizon

>B-horizon

Zone of accumulation because all the leaching materials are accumulated in horizon.

C-horizon

it is composed of broken bedrock material that is merged with underlying solid and hard bedrock material.

>Factors that influence these horizons are,

- **Erosion**
- **Wind**
- **Precipitation**
- **Human activity**
- **Topography**

3: Differentiate between Autogenic and Allugenic succession?3

>Autogenic succession

In autogenic succession the successional process

is controlled by living or biotic component of an
ecosystem
Allogenic succession
The entire successional process is controlled by
non living or abiotic components of ecosystem.

4: differniate between population and meta polutaion ?3

Metapopulation

A group of populations of same species which are separated by space. Individuals can move from one population to other. Extinction and colonization are the main processes that regulate the Metapopulation.

In biology, a **population** is all the organisms of the same group or species, which live in a particular geographical area, and have the capability of interbreeding.

5 : HAB stands for?2

HAB. **Harmful Algal Bloom**.

6: how to analysia the kind of soil ?2

7. Define food pyramid? 2)

>A food pyramid is a graphical representation of flow of energy through different trophic levels of an ecosystem.
The highest level an organism occupies on a food chain is termed as trophic level.

8. Difine species hotspotes?3)

Species hotspots
Species hotspots are geographic regions that support high level of species diversity.
Tropical rainforests are one of important species hotspots region in the world because they are home to more than 50% species.

9. population dispersion??

Above question 1

10. difference between aquatic and trestrail biomes?

Almost all the habitats found in the world can be put into two major habitats; **aquatic and terrestrial**. **Aquatic** ecosystems are found in water .**Terrestrial** animals are the animals who live on land for most or all of their life span.

11. mechanism of survival of endothermic cold and hot environment?

12. define atmosphere and role in ecosystem?

An **atmosphere** (from Greek ἀτμός (atmos), meaning "vapor", and σφαῖρα (sphaira), meaning "sphere") is a layer of gases surrounding a planet or other material body, that is held in place by the gravity of that body.

Role

The role of the atmosphere in the provision of ecosystem services.

Cooter EJ[1], Rea A, Bruins R, Schwede D, Dennis R.

Author information

Abstract

Solving the environmental problems that we are facing today requires holistic approaches to analysis and decision making that include social and economic aspects. The concept of ecosystem services, defined as the benefits people obtain from ecosystems, is one potential tool to perform such assessments. The objective of this paper is to demonstrate the need for an integrated approach that explicitly includes the contribution of atmospheric processes and functions to the quantification of air-ecosystem services. First, final and intermediate air-ecosystem services are defined. Next, an ecological production function for clean and clear air is described, and its numerical counterpart (the Community Multiscale Air Quality model) is introduced. An illustrative numerical example is developed that simulates potential changes in air-ecosystem services associated with the conversion of evergreen forest land in Mississippi, Alabama and Georgia to commercial crop land. This one-atmosphere

approach captures a broad range of service increases and decreases. Results for the forest to cropland conversion scenario suggest that although such change could lead to increased biomass (food) production services, there could also be coincident, seasonally variable decreases in clean and clear air-ecosystem services (i.e., increased levels of ozone and particulate matter) associated with increased fertilizer application. Metrics that support the quantification of these regional air-ecosystem changes require regional ecosystem production functions that fully integrate biotic as well as abiotic components of terrestrial ecosystems, and do so on finer temporal scales than are used for the assessment of most ecosystem services.

13. Keystone species and foundation species with examples.

- **Some species play a key and crucial role in maintaining the structure of a biological community. They are called Keystone species because they have a large effect on the types and abundances of other species in an ecosystem. Sharks are also a keystone species because they remove injured and sick animals from the ocean, and thus play an important ecological role.**

>Foundation species, plays a major role in shaping communities.
They create and Enhance the habitats in ways that provides benefits to other species. They facilitate the survival and reproduction of other species in the community.
>For example, elephants push over, break, or uproot trees, creating forest openings in the grasslands and woodlands of Africa.
>Foundation vs keystone species

- **A foundation species creates or provide a foundation of an ecosystem while keystone species regulates and maintains that ecosystem**

14. Generalists and specialist species.

- **Those species that have broad range of niches are known as generalist species.**

Flies, cockroaches, mice, rats, white-tailed deer, raccoons, and humans are Generalist species

- **Those species that occupy narrow range of niches**

are known as specialist species.
For example, tiger salamanders breed only in fishless ponds where their larvae will not be eaten. Some shorebirds occupy specialized niches, feeding on crustaceans, insects, and other organisms on sandy beaches and their adjoining coastal wetlands.

15. How can we analyse any kind of soil?

Soil type usually refers to the different sizes of mineral particles in a particular sample. Soil is made up in part of finely ground rock particles, grouped according to size as sand and silt in addition to clay, organic material such as decomposed plant matter.

Each component, and their size, plays an important role. For example, the largest particles, sand, determine aeration and drainage characteristics, while the tiniest, sub-microscopic clay particles and are chemically active, binding with water and plantnutrients. The ratio of these sizes determines soil type: clay, loam, clay-loam, silt-loam, and so on.



16. ~~Atmosphere and its role in ecosystem~~

17. River three zone..

Small streams join to form rivers, and rivers flow downhill to the ocean.

- **Source zone**
- **Transition zone**
- **Floodplain zone**

18. ; what permforst?

Permafrost is soil, rock or sediment that is frozen for more than two consecutive years. In areas not overlain by ice, it exists beneath a layer of soil, rock or sediment, which freezes and thaws annually and is called the "active layer".

19) Difference between plankton and phytoplankton?

- **Plankton** are drifting organisms that live in the surface layers of the ocean. They live in the top layer of the ocean, called the epipelagic zone. They are not strong enough to swim against ocean currents. The term is in contrast to nekton, who can control their movements.
- **Phytoplankton** /ˌfaɪtoʊˈplæŋktən/ are the autotrophic (self-feeding) components of the plankton community and a key part of oceans, seas and freshwater basin ecosystem. **Phytoplankton**, also known as microalgae, are similar to terrestrial plants in that they contain chlorophyll and require sunlight in order to live and grow. Most **phytoplankton** are buoyant and float in the upper part of the ocean, where sunlight penetrates the water.

20)cell theory?

- The unified **cell theory states** that: all living things are composed of one or more**cells**; the **cell** is the basic unit of life; and new **cells** arise from existing **cells**. Rudolf Virchow later made important contributions to this **theory**. ... The **cell** is the fundamental unit of structure and function in living things.

In general soils are classified into two main groups on the base of climate,

1. **Pedalfer soils**
2. **Pedocal soils**

1. Pedalfer soils

Pedalfer is composed of aluminum and iron oxides. It is a subdivision of the zonal soil ordercomprising a large group of soils in which sesquioxides increase relative to silica during soil formation. Pedalfers usually occur in humid areas. Pedalfer soils are Acidic soils formed in temperate (wet) climates and mostly are extensive leaching activity and Supports growth of deciduous forests and tall grasses.

2. Pedocal soils

Pedocal is a subdivision of the zonal soil order. It is a class of soil which forms in semiarid andarid regions. It is rich in calcium carbonate and has low soil organic matter. With only a thin A horizon (topsoil), and intermittent precipitation calcite, other soluble minerals ordinarily removed by water may build up in the B horizon (subsoil) forming a cemented layer known as caliche.Pedocal soils promotes growth of desert shrubs and small and medium grasses.

Laterite soil

A special type of Pedocal that is bright red and forms in tropical and subtropical climates with heavy rain and high temperature.

Caliche

A tough cemented layer developed by the dense mass of calcium carbonate.

21) Three properties of exotherms?

Characteristics of Ectotherms

- **Low metabolic rate**
- **Generate body heat from external environment**
- **Body insulation is very poor**

>An inguana sunning on rock to warm

22)~~population dispersion~~?

Individuals of a **population** can be spaced in different ways, called **dispersion** patterns. In uniform**dispersion**, individuals are evenly spaced. In random**dispersion**, individuals are randomly arranged. The most common spatial arrangement is clumped**dispersion**, in which individuals are clustered together.

23. Types of ecosystem

- Forest Ecosystems. Forest ecosystems are classified according to their climate type as tropical, temperate or boreal. ...
- Grassland Ecosystems. Different types of grassland ecosystems can be found in prairies, savannas and steppes. ...
- Desert Ecosystems. ...
- Tundra Ecosystems. ...
- Freshwater Ecosystems. ...
- Marine Ecosystems.

24.Name of key stone species indicator species Impact of non native species

Species that provide early warnings of damage to a community or an ecosystem are called indicator species.

For example, the presence or absence of trout species in water at temperatures within their range of tolerance is an indicator of water quality because trout need clean water with high levels of dissolved oxygen.

non native species

Species that are non native to their environment and are deliberately or accidentally introduced into an ecosystem are called non-native species.

These are also named as invasive, alien, or exotic species.

An introduced species is occasionally successful in the new environment. Introduction of non native species is a major cause of species loss.

>Species are introduced in the new Environment by two ways intentional or Unintentional.

>Non native species that have a potential to cause economic or environmental harm to humans are termed as invasive species.

In fact, most introduced and domesticated species of crops and animals, such as chickens, cattle,

and fish from around the world, are beneficial to us.

>Grass carp was introduced in Pakistan from china. It controls the growth of pond weeds as well as a great protein source.

>Non native species that have a potential to cause economic or environmental harm to humans are termed as invasive species.

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and fish from around the world, are beneficial to us.

>Grass carp was introduced in Pakistan from china. It controls the growth of pond weeds as well as a great protein source.

>However, some non-native species can threaten a community's native species and cause unintended and unexpected consequences.

>In 1957, for example, Brazil imported wild African bees to help increase honey production.

Instead, the bees displaced domestic honeybees and reduced the honey supply.

>Impact of non native species:

- **Non-native species can spread rapidly if they find a new more favourable niche.**
- **In their new niches, these species often do not face the predators and diseases they faced before.**
- **>They may be able to out-compete some native species in their new niches.**
- **Causes habitat loss and ecosystem disturbance.**

25.What is HABs?

Eutrophication

Too much nutrients causes harmful algal blooms (HABs) which are extremely toxic to living creatures (fish, shellfish, birds and

mammals).

26.Types of fresh water bodies.

include glaciers, snow, rivers, streams, lakes, wetlands, springs and aquifers and groundwater

27. Difference between hibernation and winter sleep.

True hibernation occurs in small animals in which temperature substantially falls to remain alive in cold environment. Animals remain inactive for weeks or even for months.

>Hibernation in large animals like bears is termed as winter sleep.

>Body temperature does not falls substantially and sleeping animals can wake and active quickly.

It lasts for short period of time.

Large animals have more energy reserves that enables them to survive in winter.

28. Modifications in hot and cold environment

>Survival mechanisms in hot environment

- **Sweating**
- **Panting**
- **Vasodilation**
- **Aestivation**

Survival mechanisms in cold environment

- **Fat deposits**
- **Fur**
- **Feathers**
- **Vasoconstriction**
- **Hibernation**

EXAMPLE IS wild boar

29-what factor are important in population dynamic?

Population dynamics is a study of how these characteristics of populations change in response to changes in environmental conditions.

>Such environmental conditions are

- **Temperature**
- **Presence of disease organisms or harmful chemicals**
- **Resource availability**
- **Arrival or disappearance of competing species.**

General patterns of population distribution or dispersion

1. **Clumping**
2. **Uniform dispersion**
3. **Random dispersion**

>Clumping

Individuals are found in form of groups or clusters due to uneven distribution of resources.

2. Uniform dispersion

The tendency for populations to be found evenly distributed about their habitat.

This is generally caused by a species ability to survive anywhere in their habitat. They use the resources found immediately around them, and spread out as to use all of the available resources.

3. Random dispersion

Distribution of populations in a random fashion through out their habitat.

Generally randomly distributed populations are rare while clumped distribution is common.

30-define torpor with example?

A condition in which the rate of metabolism, circulation and respiration of individuals decreases when they are not feeding. Torpor allows the individual to survive in non feeding period.

>Torpor enables these animals to survive in a condition in which they are unable to feed. It is found in bats, hummingbirds, and other animals. For example bat at a state of torpor.

31-enlist the sub field of ecology?

Sub fields of ecology are

1) Organismal ecology

2) Population ecology

3) Community ecology

4) Ecosystem ecology

5) Landscape ecology

6) Global ecology

1) Organismal ecology

Organismal ecology is the study of individual organisms' behavior, physiology, morphology, etc. in response to environmental challenges.

2) Population ecology

Population ecology is a sub-field of ecology that deals with the dynamics of species populations and how these populations interact with the environment. It is the study of how the population sizes of species change over time and space.

3) Community ecology

Community ecology is the study of the interactions between species in communities on many spatial and temporal scales, including the distribution, structure, abundance, demography, and interactions between coexisting populations

4) Ecosystem ecology

Ecosystem ecology is the integrated study of living (biotic) and non-living (abiotic) components of ecosystems and their interactions within an ecosystem framework. This science examines how ecosystems work and relates this to their components such as chemicals, bedrock, soil, plants, and animals.

5) Landscape ecology

Landscape ecology is the science of studying and improving relationships between ecological processes in the environment and particular ecosystems. This is done within a variety of landscape scales, development spatial patterns, and organizational levels of research and policy.

6) Global ecology

Global ecology is the study of the interactions among the Earth's ecosystems, land, atmosphere and oceans. Global ecology is very important because it is used to understand large scale interactions and how they influence the behavior of the entire planet, including the earth's responses to future changes.

32-enlist at least three specific sites of secondary succession occur?

Examples of secondary succession include

1. The renewal of a forest after a fire:

The fire itself destroys a majority of different types of trees and plant life. Because seeds and roots and other plant and tree parts remain in and on the soil, gradually the plants and trees begin to grow again and eventually return to the state of the original ecosystem.

2. The renewal of a crop after harvesting:
A crop is completely harvested when it becomes ripe. Without new seeds being planted, the crop can regenerate the following year due to the plants and seeds that remained after harvesting.

3. A forest renews after logging:
A large amount of trees were chopped down by loggers in order to create building materials. Over time, trees grow in and the area returns to its previous state.

Candidates for secondary succession include

- Abandoned farmland
- Burned or cut forests
- Heavily polluted streams
- Flooded lands

33-define species and enlist the primary concepts that define species?

34-define the survivel mechanism of organism in both cold and hot environment?

Warm-Blooded or Cold-Blooded?

Warm-blooded animals, which are mostly birds and mammals, need to maintain a relatively constant body temperature or they would suffer dire consequences. It doesn't matter what the outside temperature is—they must maintain the same internal temperature. For us, the commonly accepted average body temperature is 98.6 °F (even though it may vary among individuals). Most other mammals range from 97 °F to 103 °F; birds have an average body temperature of 105 °F.

Cold-blooded animals do not maintain a constant body temperature. They get their heat from the outside environment, so their body temperature fluctuates, based on external temperatures. If it is 50 °F outside, their body temperature will eventually drop to 50 °F, as well. If it rises

to 100 °F, their body temperature will reach 100 °F. Most of the rest of the animal kingdom—except birds and mammals—are cold-blooded.

Hibernation

>A condition in which animals become inactive usually in cold environments for their survival.

Animals hibernate due to following reasons,

- **Very low temperature**
- **Shortage of food**
  - **>True hibernation occurs in small animals in which temperature substantially falls to remain alive in cold environment. Animals remain inactive for weeks or even for months.**
  - **>Hibernation in large animals like bears is termed as winter sleep.**
  - **>Body temperature does not falls substantially and sleeping animals can wake and active quickly.**

Aestivation

- **>Aestivation or summer sleep is a period of inactivity and low metabolic rate in animals in response to high temperature and dry season.**
- **Aestivation is common in many invertebrates, reptiles, and amphibians.**

35. ~~atmosphere and its role.~~

~~36. Key stone and foundation species.~~

37. Reproduction potential and its effecting factors

Reproductive Potential. The number of viable offspring produced by a spawning stock is central to evaluating sustainable fishing. This **reproductive potential** is a function of spawner biomass, sex ratio, maturity at age, realised fecundity and gamete viability.

Aging, diet, environment, fertility,

~~38. How can we analyse any kind of soil~~

~~39. Generalist species and specialist species.~~

40. Define population heterophies

~~41. torpor~~

42. Survivorship curve….5

A **survivorship curve** is a graph showing the number or proportion of individuals surviving to each age for a given species or group (e.g. males or females).**Survivorship curves** can be constructed for a given cohort (a group of individuals of roughly the same age) based on a life table.

curves. Species such as humans and other large mammals, which have fewer numbers of offspring but invest much time and energy in caring for their young (K-selected species), usually have a Type I **survivorship curve**. This relatively flat**curve** reflects low juvenile mortality, with most individuals living to old age.

This basically means that most of the individuals **will** make it to adulthood but the proportion surviving into old age is greatly decreased. A type I **survivorship curve**is plotted as a convex **curve** on a graph. A type II **survivorship curve** shows a roughly constant mortality rate for the species through its entire life.

Examples include **lizards**, **hydra**, and some small **mammals**. Type III individuals initially have a rather low chance of survival. Those that do survive may live to an advanced age. Examples include many **fish** and other marine organisms.

~~43. Indicator environmnte~~

44. Role of ecological and mechanical forces in land plants.

45. Terrestrial plants are self-supportive plants, so they are strong and stiff in nature. They experience Gravity as a main environmental force while Wind as an external mechanical force.

46. Role of wind in lentic water bodies.

47. Standing bodies of freshwater that supports variety of life. Wind action is very vital in these water bodies because it increases the content of dissolve oxygen that is necessary for the aquatic life in these water bodies. A lentic water bodies includes biotic (living) plants, animals and micro-organisms, as well as abiotic (nonliving) physical and chemical interactions.

48. Use of indicator species

49. Plants or lichens sensitive to heavy metals or acids in precipitation may be indicators of air pollution. Indicator species can also reflect a unique set of environmental qualities or characteristics found in a specific place, such as a unique microclimate. However, care must be exercised in using indicator species. Judging an ecosystem based on the response of a single indicator species might be like taking a pulse on a patient and immediately prescribing a treatment without a) further examination, b) other indicators such as blood pressure, or c) knowledge of the patient's past medical history.

~~50. Species diversity~~

51. Define zooplanktons and ultraplanktons.

- **Zooplankton**, small floating or weakly swimming organisms that drift with water currents and, with phytoplankton, make up the planktonic food supply upon which almost all oceanic organisms are ultimately dependent. Many animals, from single-celled Radiolaria to the eggs or larvae of herrings, crabs, and lobsters

- **Ultra plankton**
  Very small drifting plankton having size less than 2 micrometer is called as ultraplankton. Third group consists of huge populations of much smaller plankton. These tiny photosynthetic bacteria may be responsible for 70% of the primary productivity near the ocean surface.

~~52. Define planktons and phytoplanktons~~

53. Define detrivitorose with three examples.

Detritivores, also known as detrivores, detritophages, detritus feeders, or detritus eaters, are heterotrophs that obtain nutrients by consuming detritus (decomposing plant and animal parts as well as feces).

There are many examples of detritivores; some of them include **millipedes**, **dung beetles**,**earthworms**, **fiddler crabs** and **sea cucumbers**.**Earthworms** eat decaying plant and animal matter in the soil.

54. Three reasons that decrease the size of population

Predation, migration,

~~55. Services provide by fresh water …5~~

56. Difference between feeders and filter feeders…

Filter feeders are a sub-group of **suspension feeding** animals that feed by straining **suspended** matter and food particles from water, typically by passing the water over a specialized **filtering** structure.

57. Note on soil horizon.

In a typical soil profile, four zones are exposed,

- **O-horizon**
- **A-horizon**
- **B-horizon**
- **C-horizon**

1. O-horizon

This layer generally forms above the mineral soil or occurs in an organic soil profile. The "O" stands for organic matter. It is a surface layer dominated by the presence of large amounts of organic material derived from dead plant and/or animal residues which is in varying stages of decomposition. The O

horizon is generally absent in grassland regions. The O horizon usually occurs in forested areas and is commonly referred to as the forest floor. The O horizon is a surface horizon that is comprised of organic material at various stages of decomposition. It is most prominent in forested areas where there is the accumulation of debris fallen from trees.

2. A-horizon

The A horizon is the topmost mineral horizon, often referred to as the 'topsoil'. This layer generally contains enough partially decomposed (humified) organic matter to give the soil a color darker than that of the lower horizons. The A horizons are often coarser in texture, having lost some of the finer materials by translocation to lower horizons and by erosion. The A horizon is a surface horizon that largely consists of minerals (sand, silt, and clay) and with appreciable amounts of organic matter. This horizon is predominantly the surface layer of many soils in grasslands and agricultural lands.

3. B-horizon

B horizons form below an O, A horizon and they have undergone sufficient changes during soil genesis, such that the properties of their original parent material are no longer discernible. The B horizon is commonly referred to as the "subsoil". In humid regions, B horizons are the layers of maximum accumulation of materials such as silicate clays, iron (Fe) and aluminum (Al) oxides, and organic material.

4. C-horizon

The C horizon (parent material) is below the B Horizon. This layer is little affected by soil-forming processes and they thus have a lack of pedological development. In other words, the C horizon is the unconsolidated material underlying the solum (A and B horizons). It may or may not be the same as the parent material from which the solum formed. The C horizon forms as the R horizon weathers and rocks break up into smaller particles. The C horizon is a subsurface horizon. It is the least weathered horizon. Also known as the saprolite, it is unconsolidated, loose parent material.

58. Describe photosynthesis.2 marks

During **photosynthesis**, energy from sunlight is harnessed and used to convert carbon dioxide and water into organic compounds-namely sugar molecules-and oxygen.

59. Enlist use of light 5 marks.

Role of light

60. The quality of light affects the aquatic ecosystems, the blue and red light is absorbed here and it does not penetrate deep into the water. Some algae have specialized pigments that absorbs the other colors of light.
61. The intensity of light depends upon the latitude and the season of the year. During the period from March to September the Southern Hemisphere receives less than 12 hours of sunlight while it receives more than 12 hours of sunlight during the rest of the year.
62. Some plants flower only during a certain time of the year. One of the factors is due to the length of dark period. Depending on the intensity of light the plants are classified as short-day plants (Example *Chrysanthemum sp.*, *Daturastramoniumetc*.) Long-day plants (Examples - Spinach, barley, wheat, radish, clover, etc.) Day-neutral plants (Examples - Tomato, maize, etc.)

63.what are red tides (2)

Red tide is a common name for a phenomenon known as an algal bloom (large concentrations of aquatic microorganisms) when it is caused by a few species of dinoflagellates and the bloom takes on a **red** or brown color

64.factors affecting stream flow (3)

The velocity of a stream depends on following factors

- Volume of water
- Stream channel width and depth
- Slope or gradient of stream

65. ***types of energy quality(5)***

Two types of energy on the base of energy

Quality,

1. **High quality energy**
2. **Low quality energy**

High quality energy is organized or concentrated to perform useful work.High-quality energy is concentrated form of energy and has a high capacity to do useful work.

Examples are,

- Very high-temperature heat
- Nuclear fission
- Concentrated sunlight
- High-velocity wind
- Energy released by burning natural gas, gasoline, or coal.

Low-quality energy

Low quality energy is any form of energy which is dispersed and disorderly, and has less potential or ability to be utilized for work. It is directly contrasted with high quality kinds of energy. This perception of energy quality is linked to the concept of entropy, which is a measure of how disorderly a kind of energy may be. The more disorderly, the less useful it is.The total amount of heat stored in the Atlantic Ocean is greater than the amount of high quality chemical energy stored in all the oil deposits of Saudi Arabia. But wide dispersion of this ocean's heat energy makes it unable to do useful work

66. define omnivores and give 3 exsmples? 2m

An **omnivore** is an animal that eats food from both plants and animals, which may include eggs, insects, fungi and algae. Many**omnivores** evolved to that state after many years, and are opportunistic feeders.

67. electromagnetic radiation and its 2 types? 2m

A form of kinetic energy, in which energy travels in the form of a *wave* as a result of changes in electric and magnetic fields. Human eye can detect electromagnetic radiations of wavelength range 380-760nm.

>On the base of wavelength there are two types of electromagnetic radiations.

1. **Electromagnetic radiations with short wavelength**
2. **Electromagnetic radiations with long wavelength**

68.

Q5 how human influence on environment, enlist variou influences, 5m

Human influence reaches even the farthest corners of the globe; species in the Arctic and Antarctic are contaminated by pollutants created thousands of miles away and carried through the atmosphere.

We are modifying the functioning of the entire planet, changing the earth's atmosphere through,

- Industrial release of carbon dioxide
- Diminishing the ozone layer by the production of chlorofluorocarbons.

In general, loss of biodiversity in a region may lead to,

- Decline in plant production.
- Lowered resistance to environmental distresses such as drought.
- Increased variability in certain ecosystem processes such as plant productivity, water use, and pest and disease cycles.

69.***Ectoderm and their characters***

>Organisms that control their body temperature by means of external source.

They are on the mercy of environmental temperature.

Their body temperature fluctuates with respect to external environmental temperature.

>Characteristics of Ectotherms

- **Low metabolic rate**
- **Generate body heat from external environment**
- **Body insulation is very poor**

>An inguana sunning on rock to warm

66. 3=How decomposer work

Decomposers break down dead organisms and feces by digesting organic matter

extracellularly and absorbing the products of digestion

67. **Enlist 3 specific sites where recovery succession occurs**.

Succession benefits: o Development of an ecosystem. o Sustainability of ecosystem. o Recovery of ecosystem from damage caused by humans or any other natural disaster. o Balance of nature

68. 4.Write density dependent factors

mcqs

1: Number of components of Ecology? 6

2: The species occupation in environment ?

3: coral mills used songs Birds as ?

4: how many types of Ecological pyramids ?

5: lotic system example ?

Springs

6: in which Zone of Soil is leaching present ?

7:hibernation defination

Low metabolic rate

8: marain ecosystem is the great source of ?
Biodiversity

9: magnetic radition limtis which an eye can see ?

10 : lichen and mosses are example of ?
Primary sucession

11: florjid founds in
I guess in norway

12: organisim which feed on large pieces of dead body are ?
Scavenger

13: mechanical force influenced in the terrestrial palnt most ?
14 :75 ponds food eat by which animal ?
Griffea

15:best example of scavnger
^vulture
^hawk
^recentiods
^all

16:peocess of emigration cause ?
^dcresee in popution or incres in popution

17 : population diversity of large animal are
^ low
^ high
^low high
^ remain constant
18: procedure involving in forming the soil
Wheathering
19 : current human popution in the world ?
20: indictor species definition

Mcqs:1)The term ecology was invented by the Ernst Haeckel in.
Ans: 1966
2) Trophic means??
Ans: food
3) In terrestrial ecosystem plant capture --- solar energy falls on their levees, to make food..
Ans: 1% I think…m 1 py lgaya..
4) Diversity which reflects the biological complexity of an ecosystem.
Ans: functional ecosystem.
5)The abyssal zone has temperatures around.

Ans: 3 to 4c° I think..
6) The topmost mineral horizon, often referred to as the topsoil.
Ans: A horizon..
7) Which fat gives energy to animal's brain and heart ?
Ans: brown fat
8)Lentic water bodies posses how many zones ?4

Mcqs Human eya can detect wavelength of radiation? Endotherm are organisms which can? Specialist species have range? Gernalist specise have range? Narrow ya wide btana tha Vultur is example of ? Scavangers are animals which feed on? Lichen is example of? Species is ? Agy definatnz mai sy right slct krni thi isi trha population ki b Taxis is response ti ? Biotic abiotic factr btana th

1The plant name viola arvensis is present in.....
2number of cockroach species in the world....3500
3species vich r intentionally recognized in to an ecosystem are called as....
4 the process that initiates the food chain.....
5 which 1 of the following is omnivore.....
6which of the following can eat both plants n animals....
7 which one of the following is imp in sustainability & protection of environment

Objective
Cell differentiated on the basis of ,,,,,,,,,,,,,
No of cockroach are,,,,,,,,
Human eye can see light wavelength of …….
Light independent factor is……
In o-horizone……..is present
Leaching accur in …….
Define lakes….
Lotic zone example is……
Energy flow in …….
example of secondry succession is ,